

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

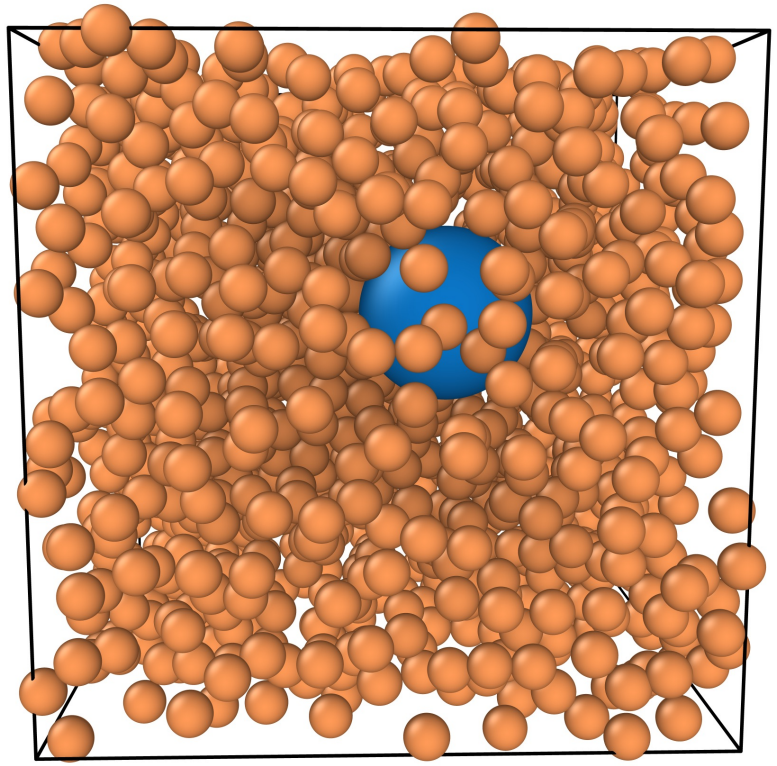
Learning Solvent Properties from Tracer Particle Diffusion

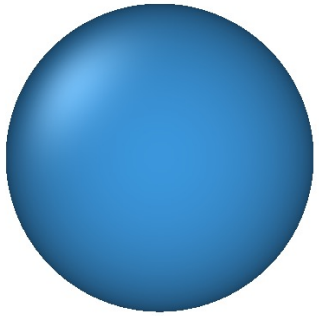
Bahadir Rusen Argun, Kaan Kirmanoglu

ME/CS 598 SciML

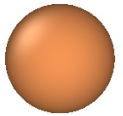
11/28/2022

System of Interest – Colloid Diffusion





Colloid Particle
Radius = 4r

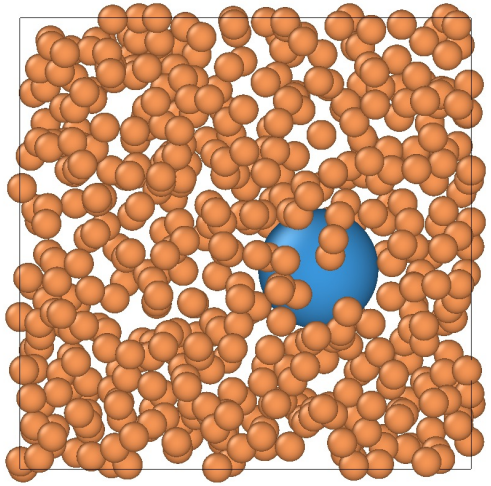


Solvent Particle
Radius = r

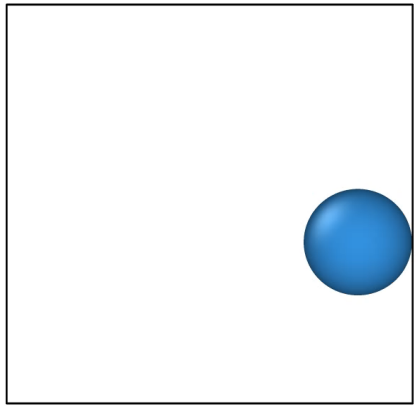
Molecular Dynamics Simulation :

- All particles interact with core repulsion
- Periodic boundary conditions
- Colloid particle will randomly move (Brownian motion)
- Density (ρ) and temperature (T) of the solvent

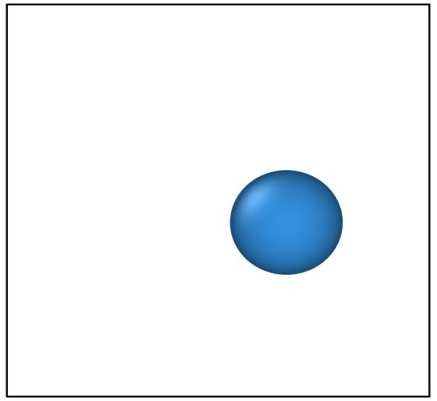
Machine Learning Task 1



Hot Solvent
T = 2.5



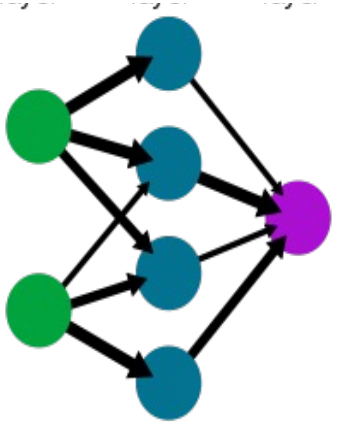
Cold Solvent
T = 0.3



Many trajectories at different temperatures

Predictor Variable $traj_i$

```
#  x      y      z
0.000 0.000 0.000
-0.069 0.068 -0.054
-0.173 0.076 -0.100
-0.270 0.047 -0.168
-0.333 0.000 -0.272
...
```



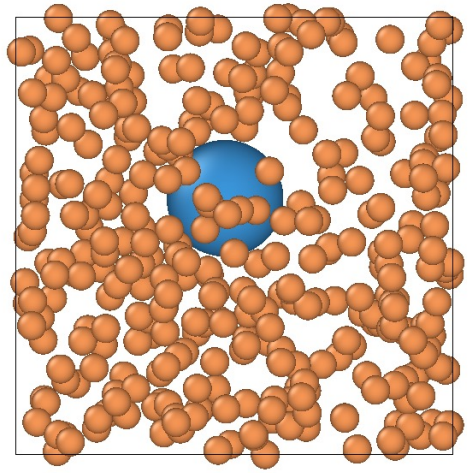
Target Variable

Temperature = T_i

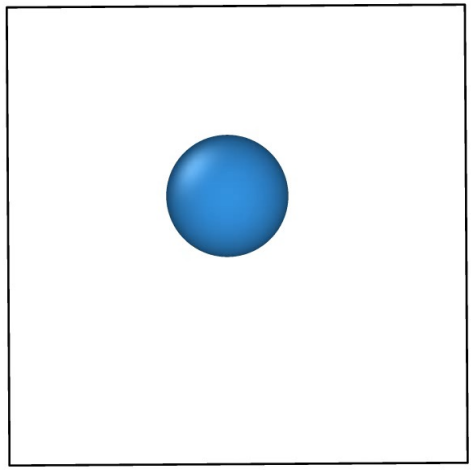
Predict solvent temperature from the trajectory of the colloid

Machine Learning Task 2

Low density solvent

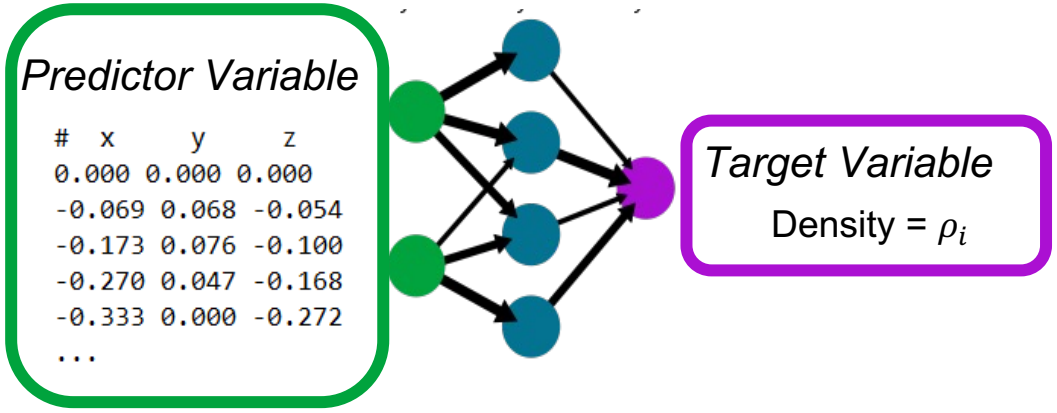


$\rho = 0.1$

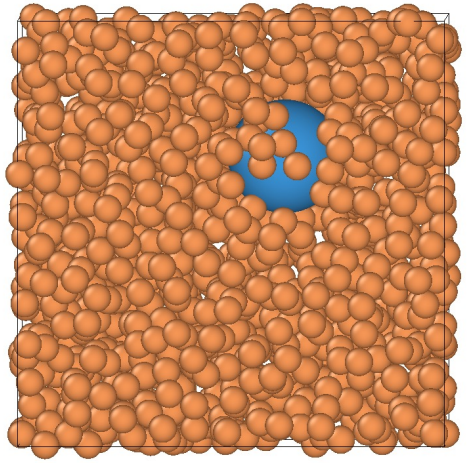


Predict solvent density from the trajectory of the colloid

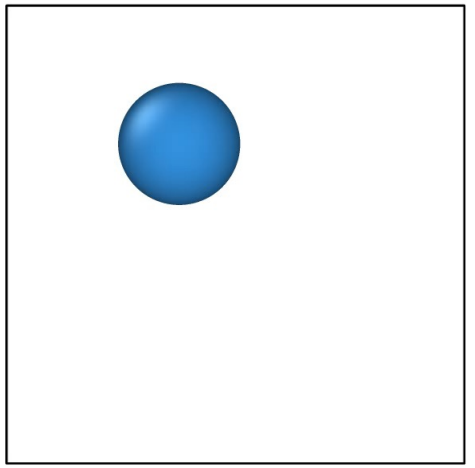
Density : $\rho = \frac{N_{solvent}}{Box\ Volume}$



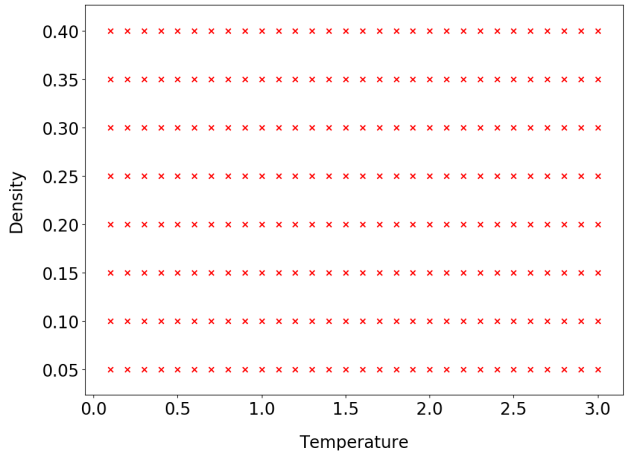
High density solvent



$\rho = 0.3$



Data and Network Structure

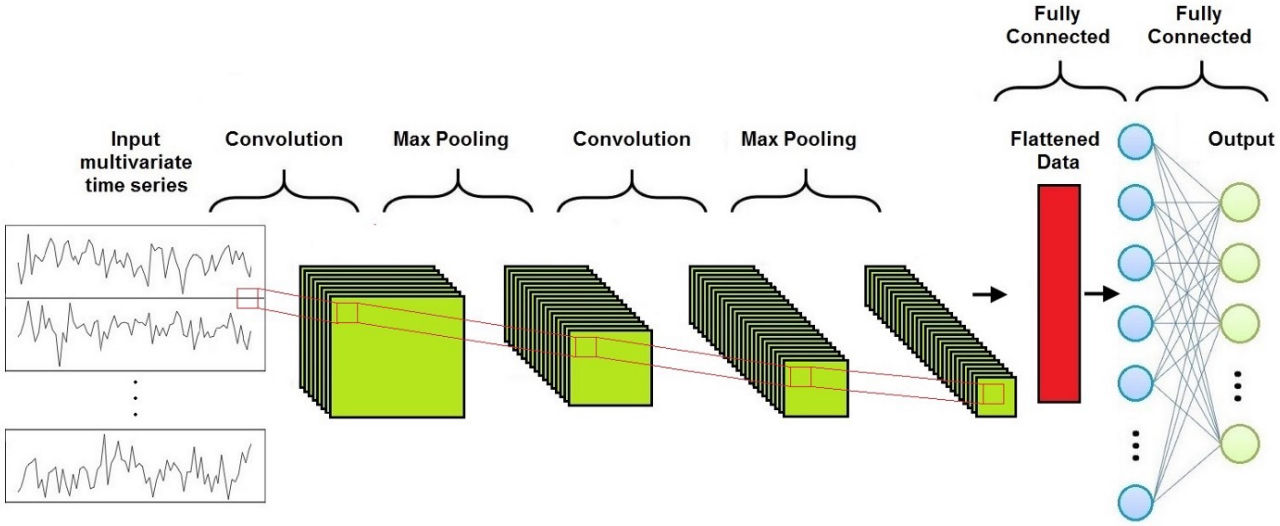


- Simulation data generated at 240 different (T, ρ) pairs

- 20 GBs of data used for training and testing
- %80 training - %20 testing

~ $1.7e9$ timesteps in total

~ 4500 independent trajectories for each (T, ρ) pair



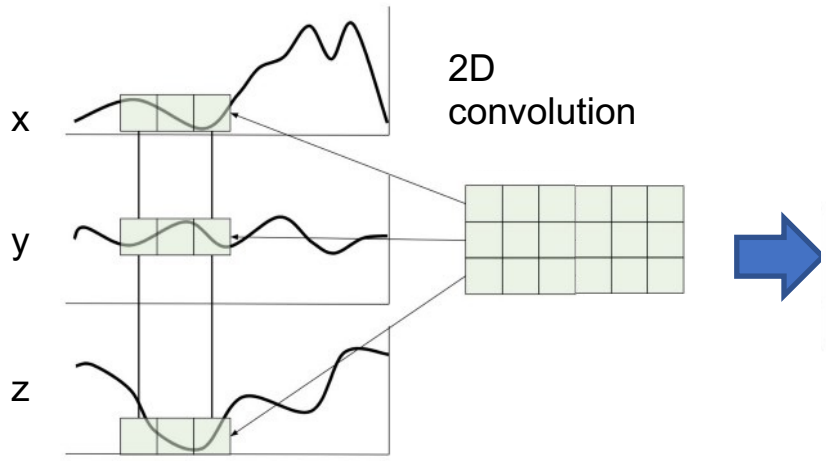
Convolutional Layer : A Linear Layer that slides over the input

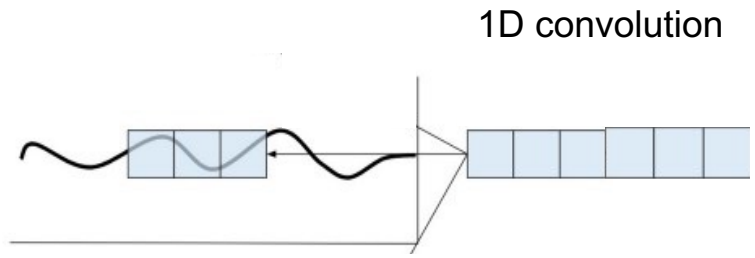
Max Pooling : Used with convolutional layers, provides translational invariance for features detected by convo layers

Zhao, Bendong, et al. "Convolutional neural networks for time series classification." Journal of Systems Engineering and Electronics 28.1 (2017): 162-169.
https://towardsdatascience.com/how-to-use-deep-learning-for-time-series-data-f641b1b41a96

Network Structure 2

```
#  x      y      z
0.000 0.000 0.000
-0.069 0.068 -0.054
-0.173 0.076 -0.100
-0.270 0.047 -0.168
-0.333 0.000 -0.272
...
```





Loss function: Mean Square Error
Batch size: 256

For Density

```
self.convo1 = nn.Conv2d(1,1,(40,3),stride=1,bias=False)
self.relu1 = nn.ReLU()
self.pool1 = nn.MaxPool1d(kernel_size=2)
self.convo2 = nn.Conv1d(1,1,4,stride=1,bias=False)
self.relu2 = nn.ReLU()
self.pool2 = nn.MaxPool1d(kernel_size=2)
self.linear1 =  nn.Linear(750, 500)
self.relu3 = nn.ReLU()
self.linear2 = nn.Linear(500, 1)
```

2 Convo layers with kernel size : 40 4
Linear layer width : 500
Input trajectory length : **800 timesteps**
Learning rate : 0.0005 (Adam)

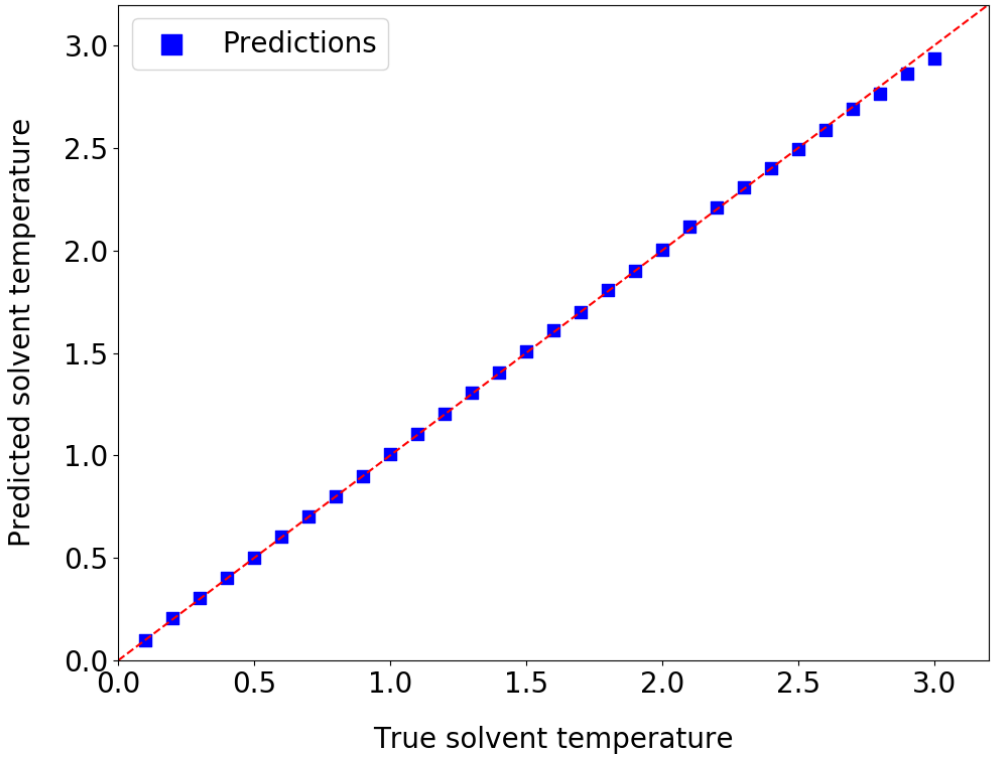
For Temperature

```
self.convo1 = nn.Conv2d(1,1,(4,3),stride=1,bias=False)
self.relu1 = nn.ReLU()
self.pool1 = nn.MaxPool1d(kernel_size=2)
self.convo2 = nn.Conv1d(1,1,40,stride=1,bias=False)
self.relu2 = nn.ReLU()
self.pool2 = nn.MaxPool1d(kernel_size=2)
self.convo3 = nn.Conv1d(1,1,40,stride=1,bias=False)
self.relu3 = nn.ReLU()
self.pool3 = nn.MaxPool1d(kernel_size=2)
self.linear1 =  nn.Linear(1220, 100)
self.relu3 = nn.ReLU()
self.linear2 = nn.Linear(100, 1)
```

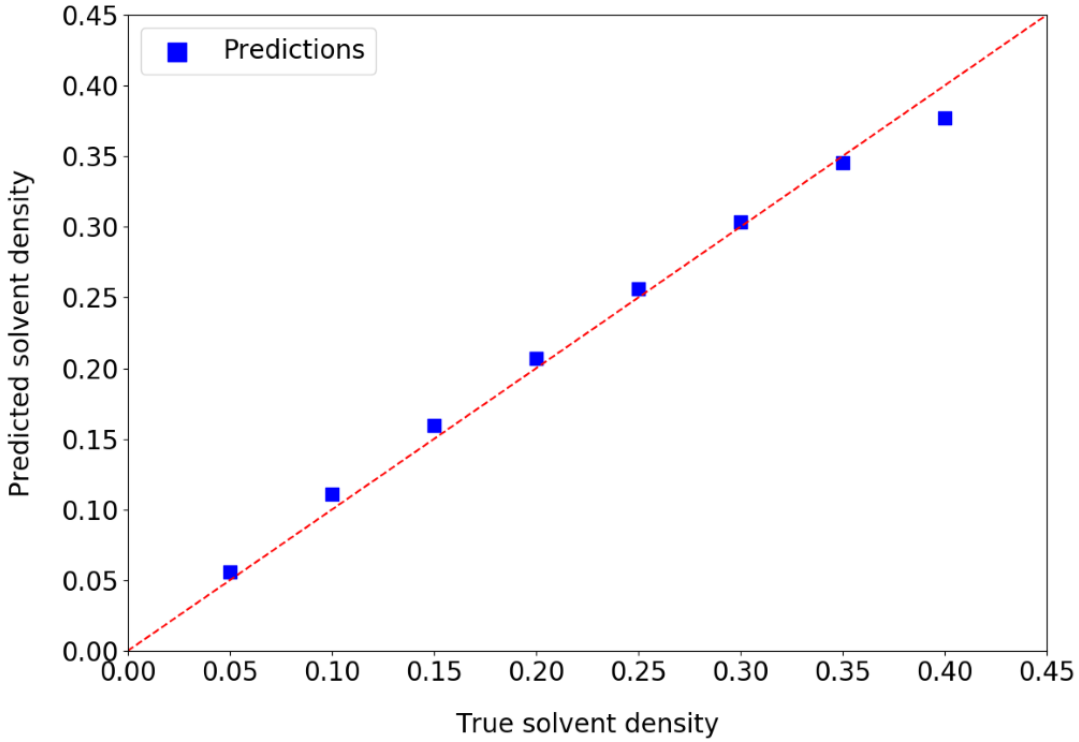
3 Convo layers with kernel size : 4 40 40
Linear layer width : 100
Input trajectory length : **10000 timesteps**
Learning rate : 0.0003 (Adam)

Results

T test results for $\rho \geq 0.15$



ρ test results



Machine Learning Task 3

Predict diffusion constant of the tracer colloid from solvent properties (temperature, density)

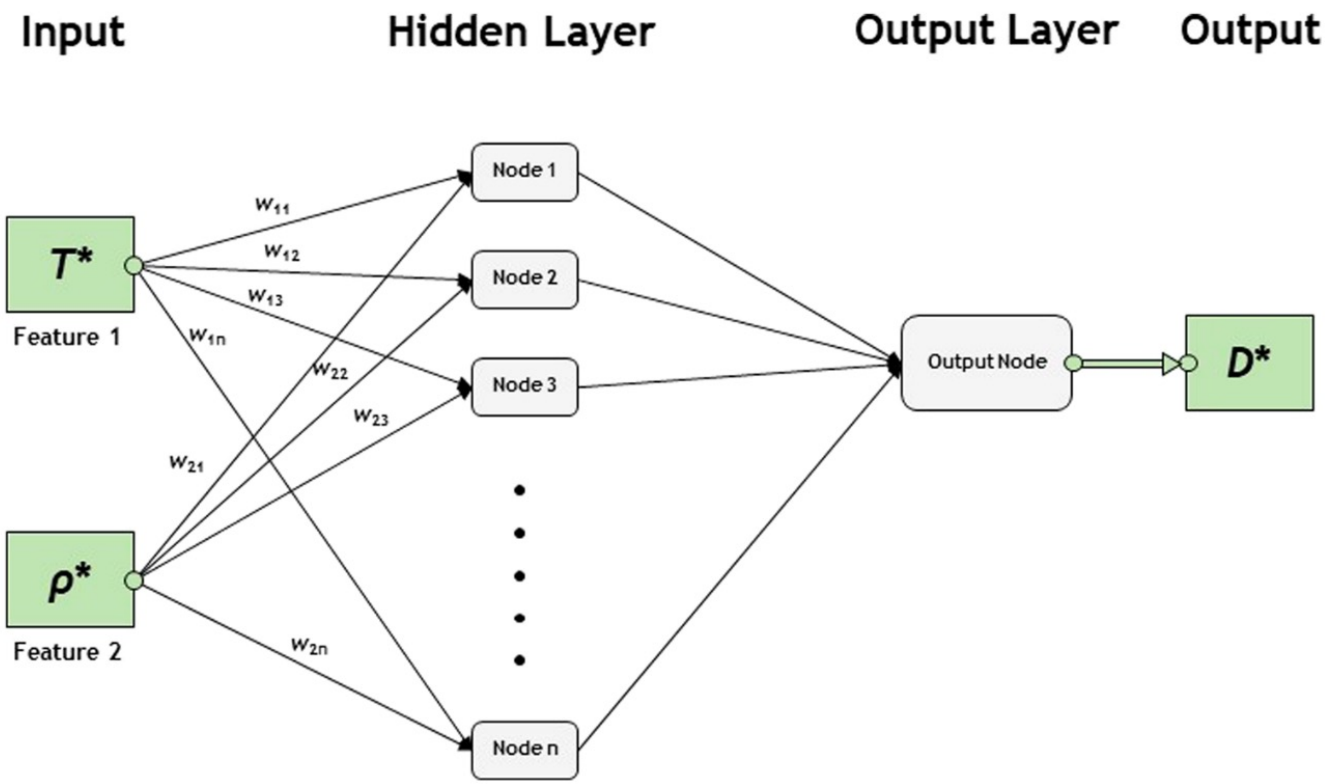


FIG. 3. Schematic of the artificial neural network (ANN) process, showing only a single hidden layer with *n* nodes.

Allers, J. P., Harvey, J. A., Garzon, F. H., & Alam, T. M. (2020). Machine learning prediction of self-diffusion in Lennard-Jones fluids. *Journal of Chemical Physics*, *153*(3). https://doi.org/10.1063/5.0011512

MD post-processing/ NN pre-processing

MD (hoomd)

Trajectories, $x(t)$

Post/Pre-Processing

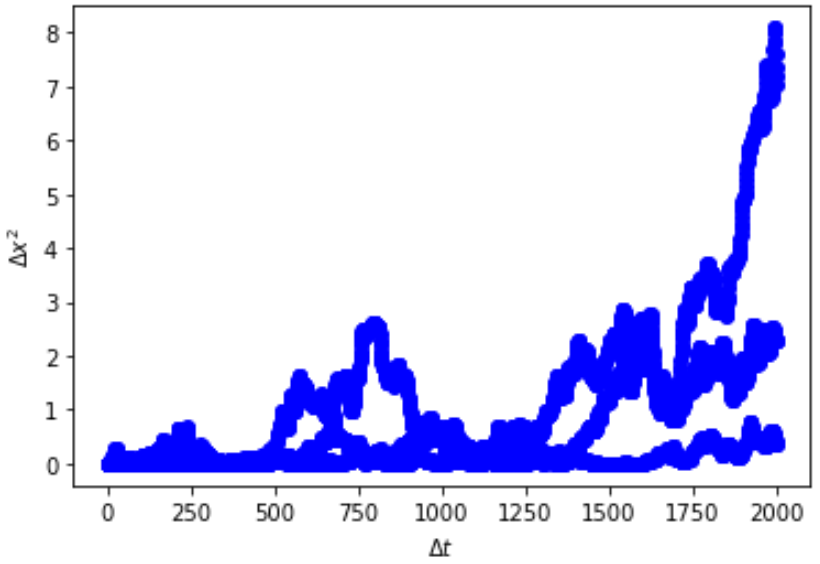
$x(t) \rightarrow D(\rho^*, T^*)$

$NN \rightarrow D_{pred}$

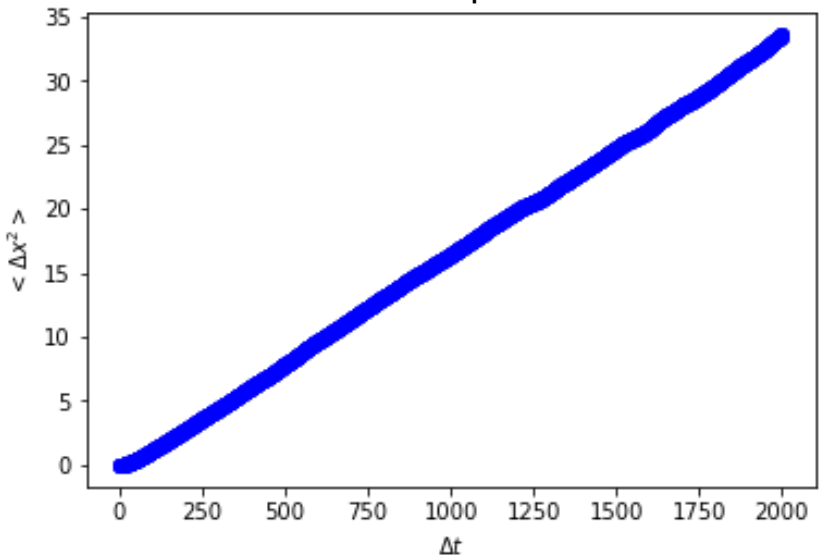
Colloids trajectories obey Brownian Motion:

$$\langle \Delta x^2 \rangle = 2Dt$$

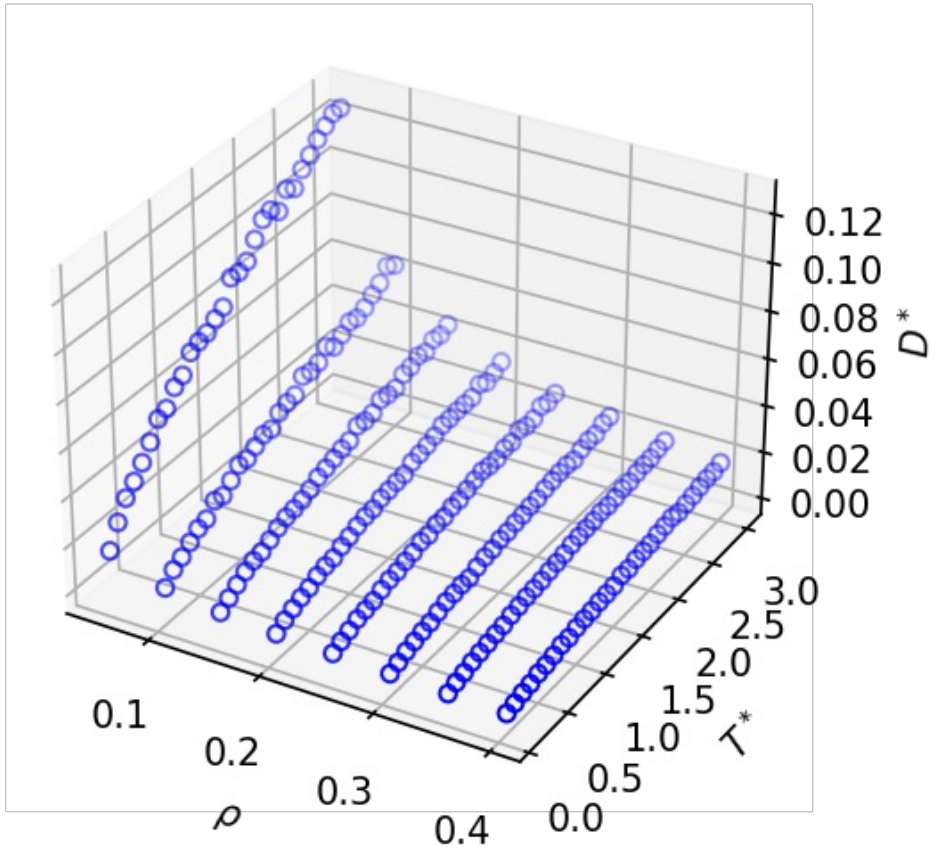
$\rho^* = 0.30, kT^* = 0.10$



2700 samples



- Data from 240 ρ^*, T^* combination



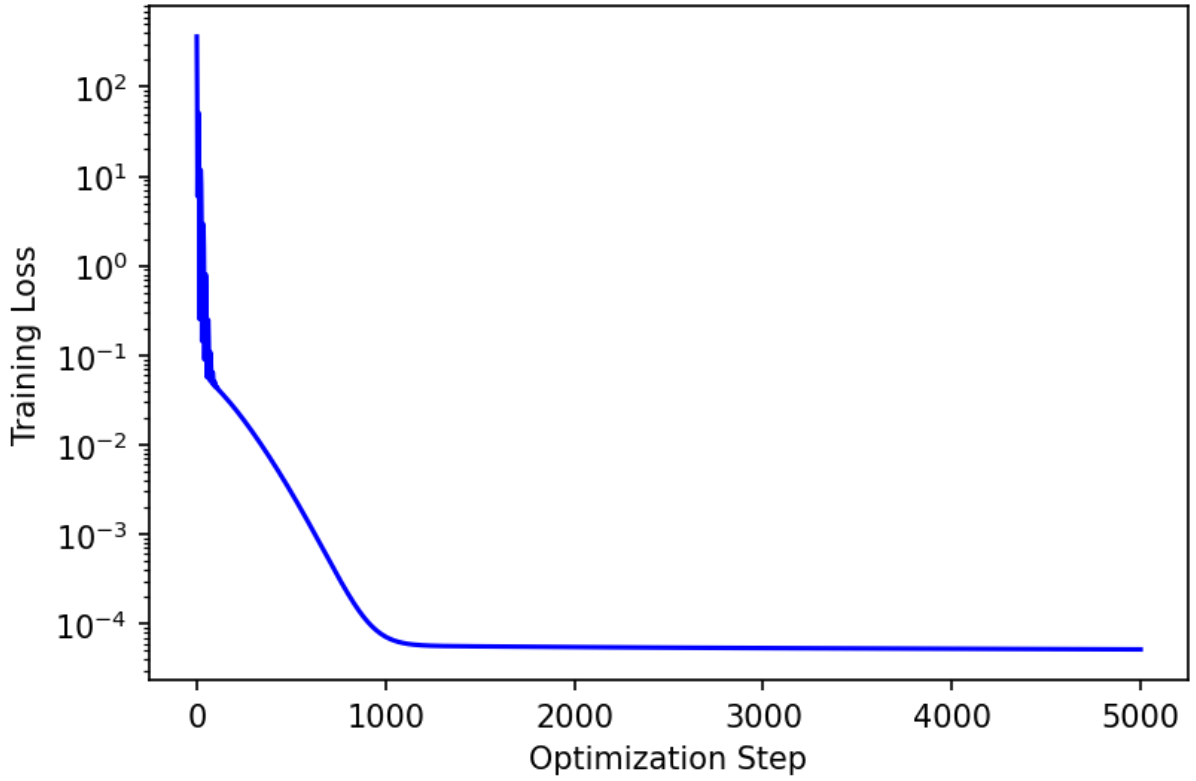
Task 3.1 $D^*(T^*)$

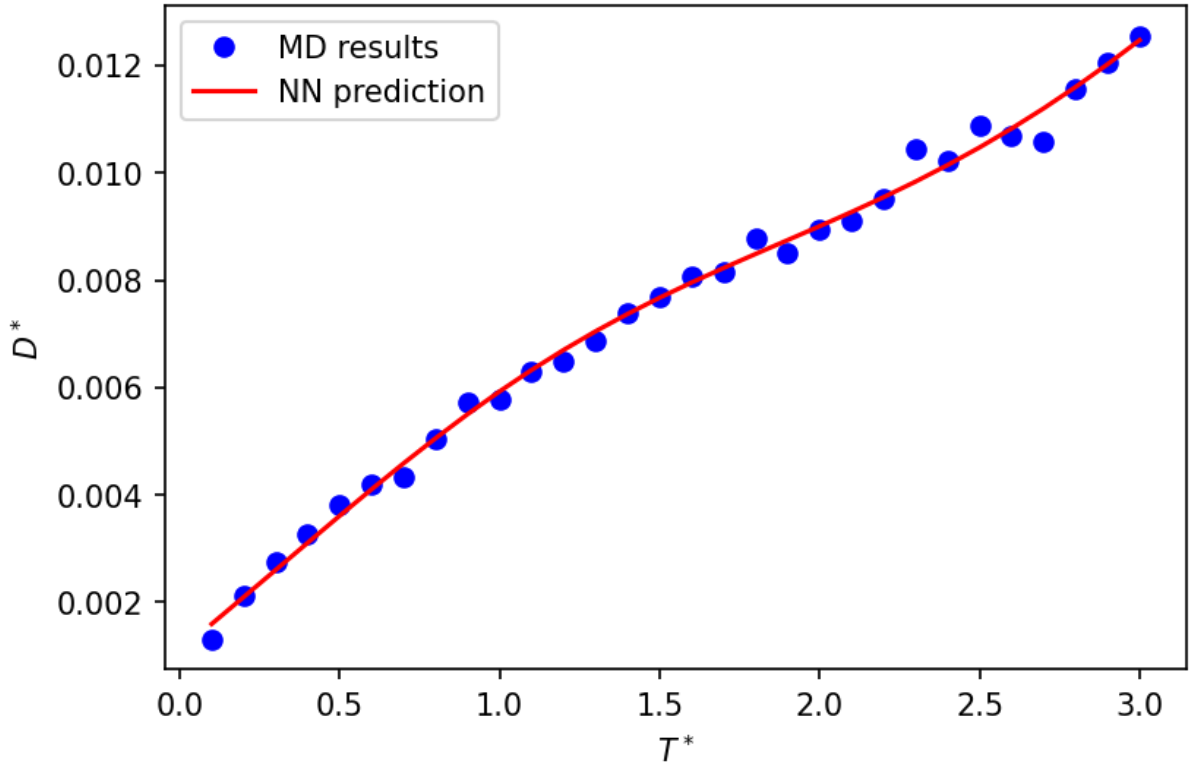
- nn_width=10
- activation = nn.Tanh()
- Loss_weigth = 1000

```
lr = 0.01
n_epochs = 5000
opt = torch.optim.Adam(model.parameters(), lr)
```

```
def loss_function(model,x):

    d_p = model(x.float())
    loss = torch.pow(D_true - d_p, 2).mean()
    return 1000*loss
```





Task 3.2 $D^*(T^*, \rho^*)$

- nn_width=10
- activation = nn.Tanh()
- Loss_weigth = 1000

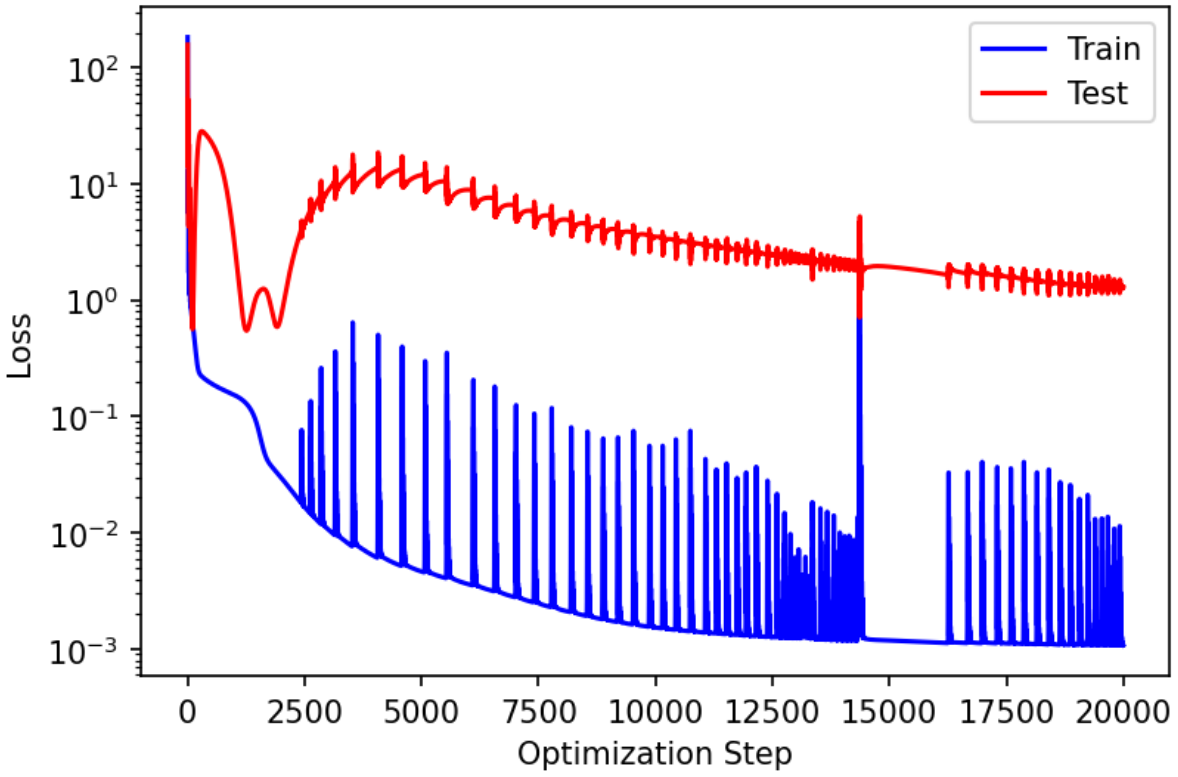
```
model = ffnn()
lr = 0.01
n_epochs = 20000
opt = torch.optim.Adam(model.parameters(), lr)
```

```
def loss_train(model,T, rho):

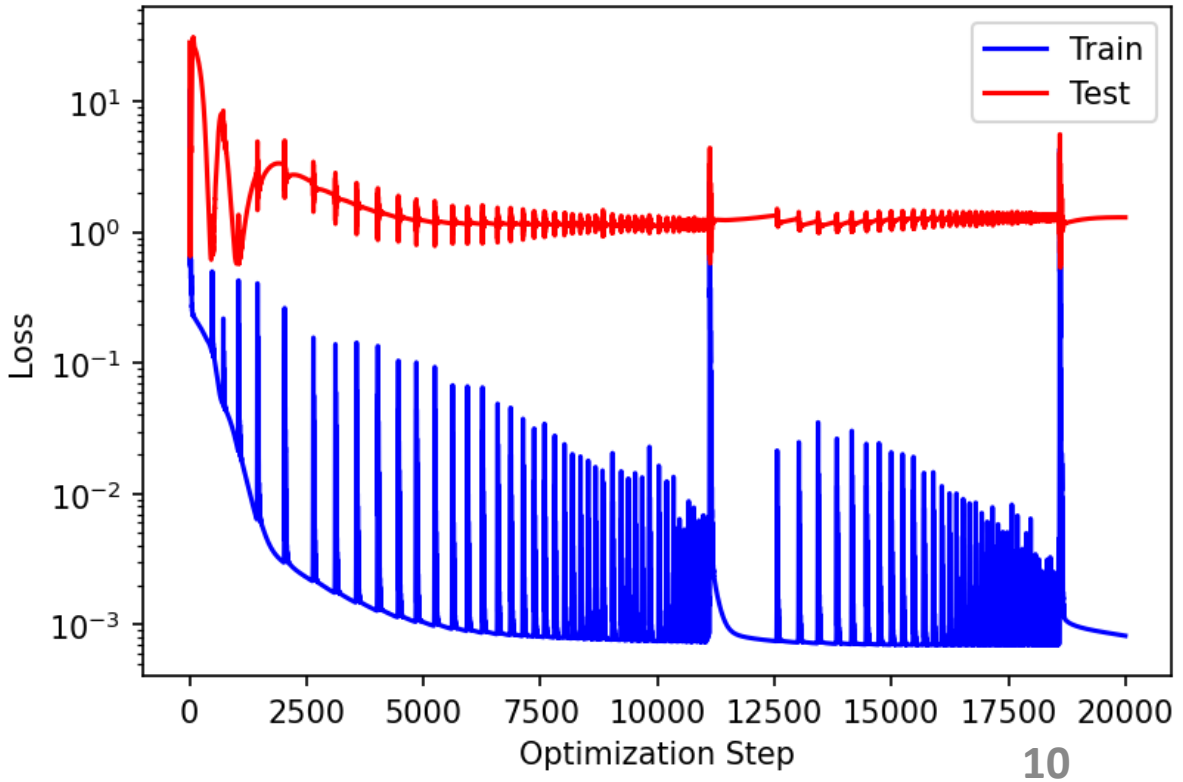
    d_p = model(T.float(),rho.float())
    loss = torch.pow(D_train - d_p, 2).mean()
    return 1000*loss

def loss_test(model, rho, T):
    d_p = model(T.float(),rho.float())
    loss = torch.pow(D_test - d_p, 2).mean()
    return 1000*loss
```

Num_hidden = 1

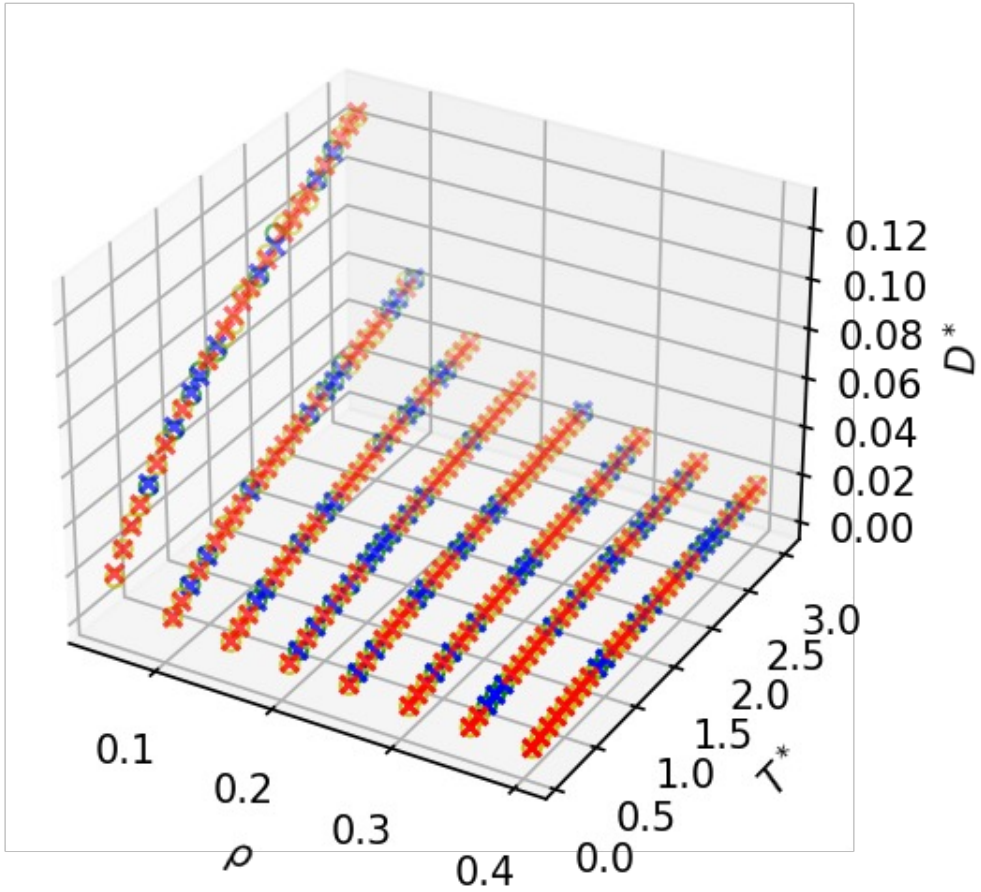


Num_hidden = 2

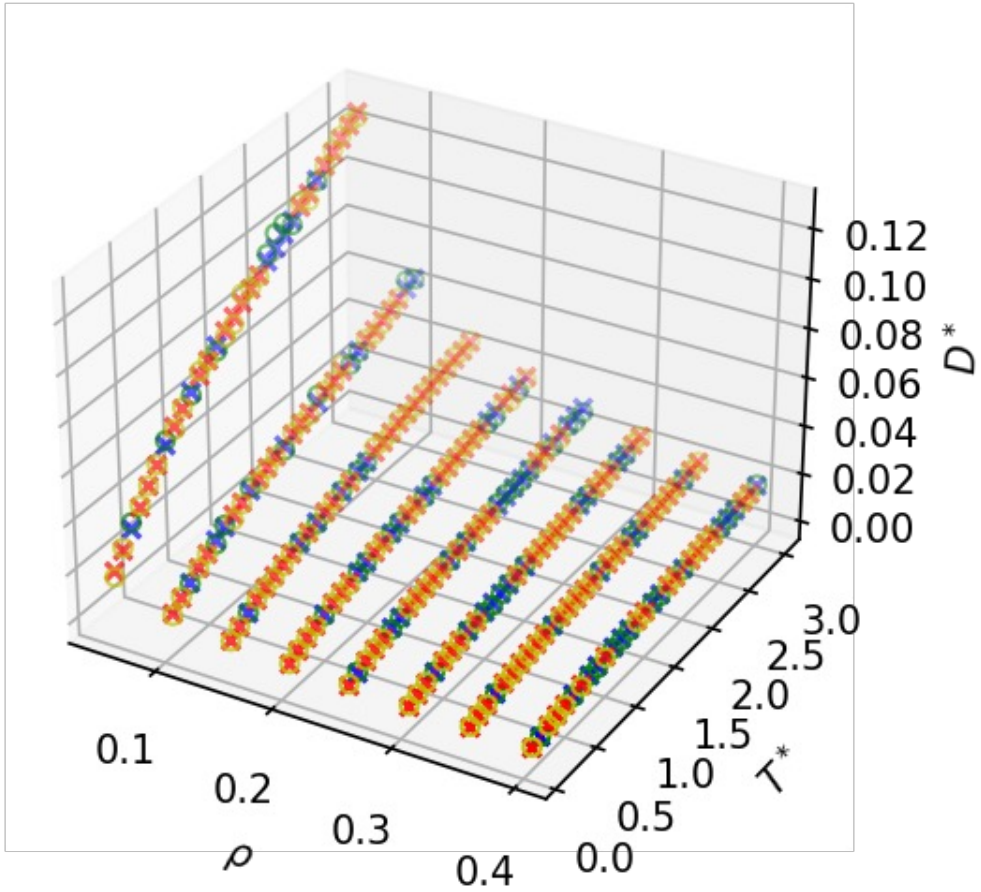


Training Data

Num_hidden = 1



Num_hidden = 2

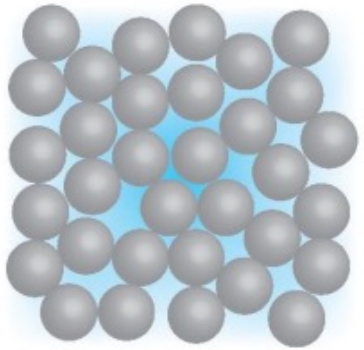


Challenges and Limitations

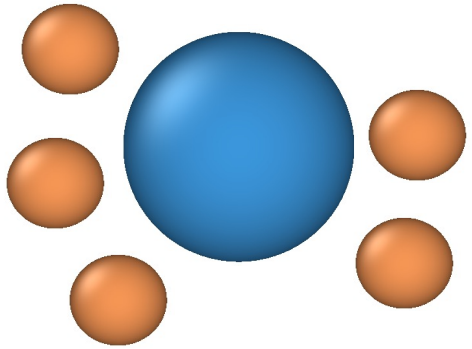
- Large amount of data: Needs large memory
- Training is very sensitive to hyperparameters: Batch size, learning rate
- Testing/Training sensitive to network structure
- Need a better method to initialize network

- Can't learn density and temperature with the same network
- Can't learn temperature at low densities
- Can't learn temperature if input trajectories have a wide range of densities
- Practically relevant density range not covered : $\rho_{water} \sim 0.55$

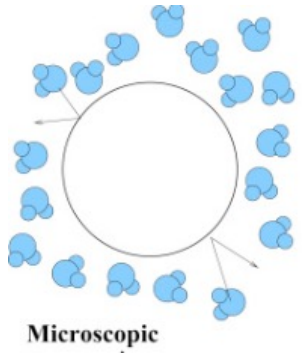
Theory available when:
$r_{colloid} \cong r_{solvent}$



We're at : $r_{colloid} = 4r_{solvent}$



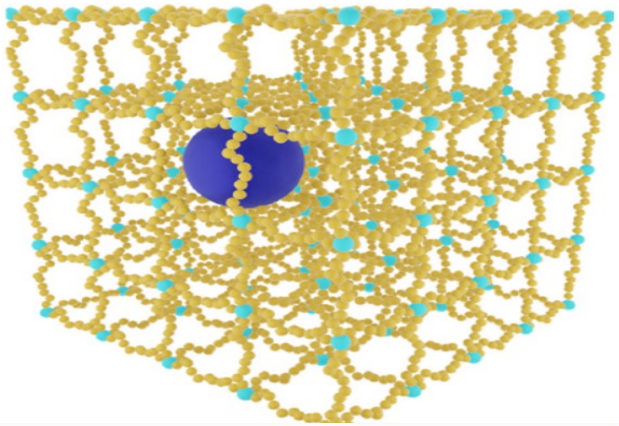
Theory available when:
$r_{colloid} \gg r_{solvent}$



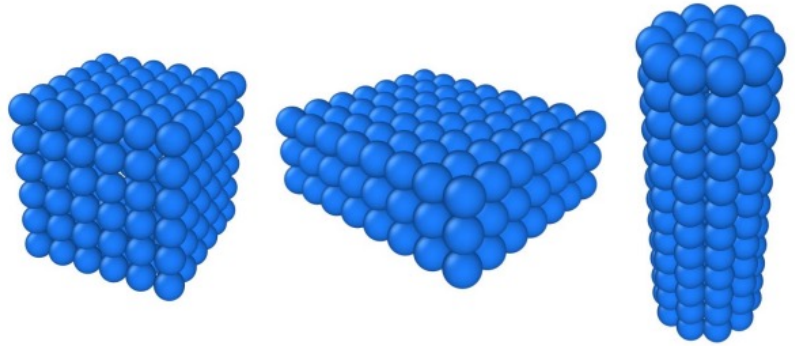
Manoharan, Vinothan N. "Colloidal matter: Packing, geometry, and entropy." Science 349.6251 (2015): 1253751.
Huber, Gary A., and J. Andrew McCammon. "Brownian dynamics simulations of biological molecules." Trends in chemistry 1.8 (2019): 727-738.

Possible Future Directions

- Learn the **non-spherical shape** of a colloid from tracer particle trajectory: Plate, cube, rod… And predict diffusion constant as shape with third parameter.

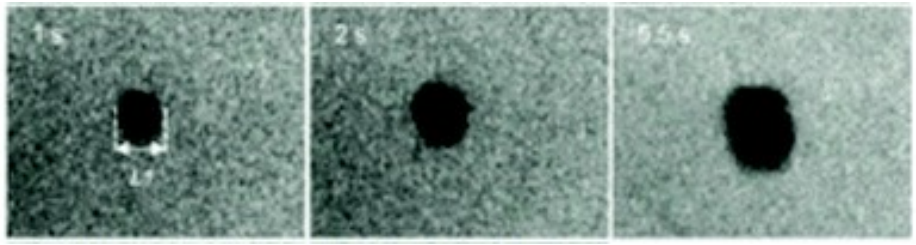


- Learn **polymer network** properties from tracer particle trajectory : Chain length distribution, crosslinker density …



- Learn **polymer melt** properties from tracer trajectory

- Use **experimental trajectories**: Can networks trained with simulations facilitate learning from experiments?



Lu, Yu, Xin-Yue Liu, and Guo-Hui Hu. "Double-Spring Model for Nanoparticle Diffusion in a Polymer Network." Macromolecules (2022).
Smith, John W., and Qian Chen. "Liquid-phase electron microscopy imaging of cellular and biomolecular systems." Journal of Materials Chemistry B 8.37 (2020): 8490-8506.

Thank You
Questions?